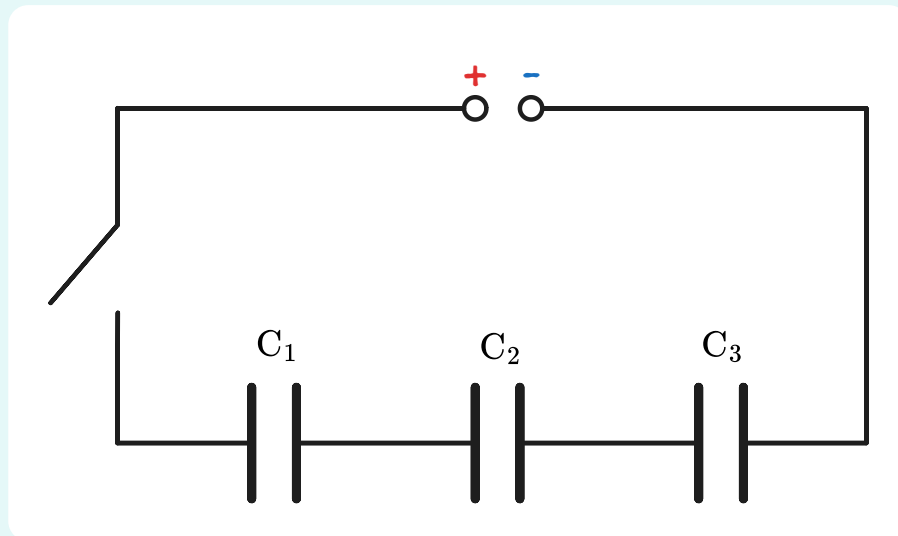


🔗 Deep Dive: Electrostatic Induction

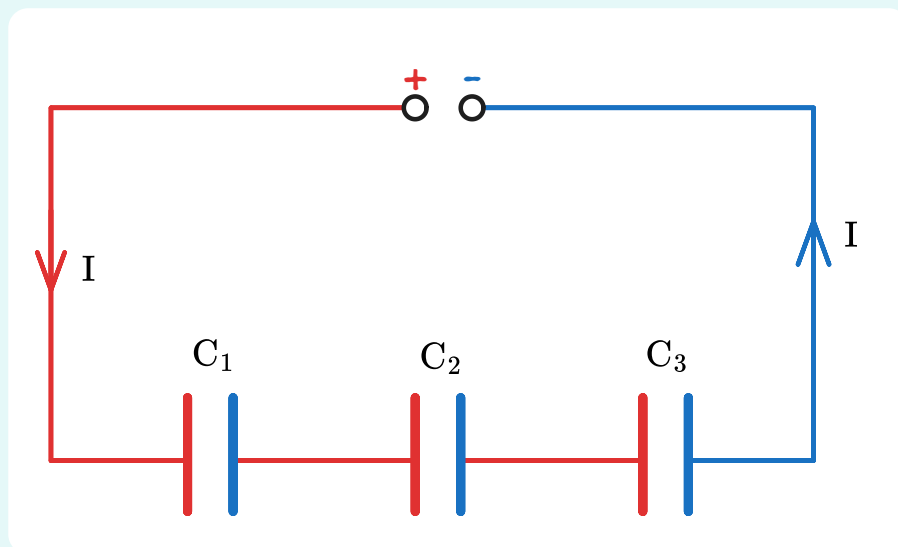
For a circuit with many capacitors, the circuit only "sees" the $+Q$ charge at the very start and the $-Q$ at the very end, the internal charges cancel out.

Consider the following circuit:



- When the switch is closed, the capacitor plates are all uncharged.

Imagine you close the switch:



When the switch is closed:

1. Current flows from positive terminal of power supply to negative terminal.
2. Let's assume that in a time t a charge $+Q$ flows through the circuit.
3. The charge of $+Q$ reaches the left plate of C_1 , this positive charge repels $+Q$ of charge from the right plate of C_1 .
4. The repelled $+Q$ flows to the left plate of C_2 .

5. This process repeats until it reaches the last capacitor.